

Business

Requirement Document



Overview

Project Title	MediConnect – Online Hospital Appointment Management System.
Project Manager	Akib Rahman.
Status	Under Review

Executive summary

ABC Hospital currently relies on manual appointment scheduling through phone calls and in-person visits. This process often results in scheduling errors, longer waiting times, inefficient resource utilization, and poor patient experience. To address these challenges, the hospital plans to implement an Online Hospital Appointment Management System (MediConnect). The proposed solution will allow patients to search doctors, view available appointment slots, book appointments online, make payments, and receive appointment confirmations automatically.

Project objectives

- Reduce appointment booking time by 70%
- Improve patient satisfaction Reduce manual workload for receptionists
- Increase online appointment adoption

Details

Project Scope

Scope Breakdown	Details
Patient registration & Login	Customers can register or login in the app.
Doctor Search	Customers can search desired / available doctor.
Appointment booking	Customers can book desired appointment.
Appointment Cancellation	Customers can cancel their appointment if they wish.
Online payment	Multiple online payment options.
SMS/email notifications	Get notifications on important updates.

Key stakeholders

- Patients / Customers
- Doctors
- Receptionists
- Hospital Managers
- System Administrators
- Admins

SWOT Analysis

Strength	Weakness
• Reduces manual appointment scheduling and paperwork. • Provides 24/7 online appointment booking for patients. • Minimizes scheduling conflicts and double bookings. • Improves operational efficiency and reduces receptionist workload.	• Initial implementation and infrastructure costs may be high. • Staff will require training to use the new system effectively. • Some patients, especially elderly users, may have difficulty using the system
Opportunities	Threats
• Introduce telemedicine and virtual consultations in future releases. • Develop Android and iOS mobile applications.	• Cybersecurity threats, including unauthorized access and data. • breaches. Low adoption if users prefer traditional booking methods.

Business Requirements

- System shall allow patient registration.
- System shall allow secure login.
- System shall display doctor schedules.
- System shall allow appointment booking.
- System shall process payments.
- System shall allow cancellation.
- System shall send notifications.
- System shall send notifications.
- System shall maintain appointment history.

Functional Requirements

- User Registration
- User Login
- Doctor Search
- View Availability
- Appointment Booking
- Appointment Cancellation
- Online Payment
- Appointment History
- Email Notification
- SMS Notification
- Report Generation
- Dashboard View

Non-Functional Requirements

- Response time under 3 seconds
- 99.5% uptime
- Secure password encryption
- Mobile responsive
- Support 5,000 concurrent users
- Daily data backup
- Audit logging

User Stories

- As a patient, I want to register an account so that I can book appointments online.
- As a patient, I want to search doctors by specialization so that I can find the appropriate doctor.

- As a patient, I want to view available time slots so that I can choose a convenient appointment time.
- As a patient, I want to cancel appointments so that I can manage my schedule.
- As an administrator, I want to generate reports so that I can monitor hospital performance.

Risks

Risk	Impact
User Resistance	Medium
Payment Gateway Failure	High
Server Downtown	High
Data Breaching	Critical
Incorrect Scheduling	High

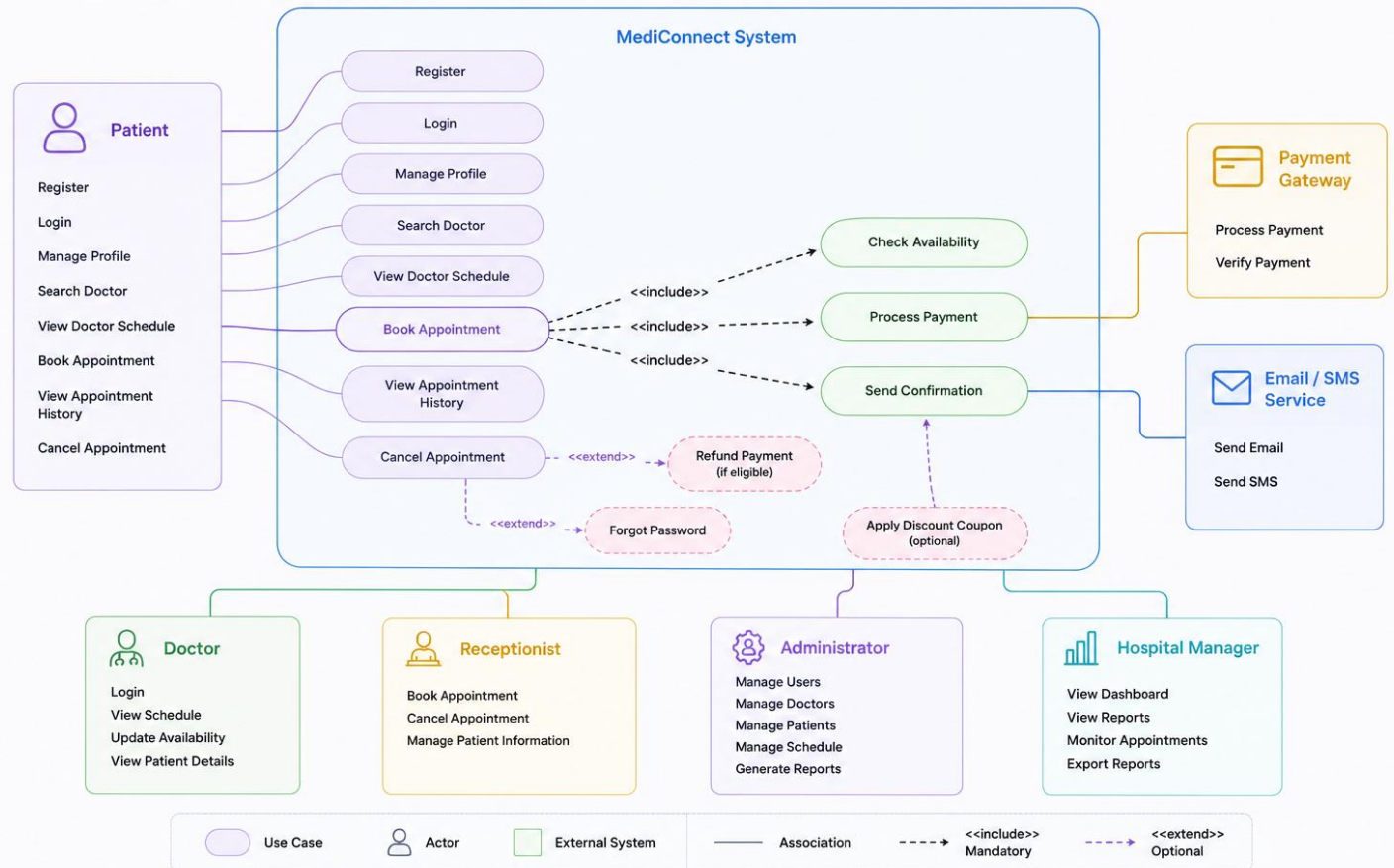
Success Metrics

KPI	Target
Online Bookings	80%
User Satisfactions	>90%
Appointment Conflicts	<2%
Booking completion rate	>85%

Use Case Diagram

Use Case Diagram – MediConnect

Online Hospital Appointment Management System



Product Backlogs

Epic	Product backlog item	Priority
User management	User registration	High
User management	User login	High
User management	Forgot password	High

User management	Reset password via email	High
User management	User profile management	High
User management	Change password	Medium
Doctor management	View doctor list	High
Doctor management	Search doctor by name	High
Doctor management	Filter by specialization	High
Doctor management	View doctor profile	Medium
Appointment	View available time slots	High
Appointment	Book appointment	High
Appointment	Cancel appointment	High
Appointment	Reschedule appointment	High
Appointment	View appointment history	Medium
Appointment	Receive booking confirmation	High
Appointment	Receive appointment reminder	Medium
Appointment	View upcoming appointments	Medium
Payment	Online payment	High
Payment	Payment status verification	High
Payment	Download payment receipt	Medium
Payment	Refund processing	Medium
Notification	Email notification	High
Notification	Sms notification	High
Notification	Push notification	Low
Doctor portal	Doctor login	High

Doctor portal	View daily schedule	High
Doctor portal	Update availability	High
Doctor portal	View patient details	Medium
Reception	Receptionist booking support	Medium
Reception	Manage patient records	Medium
Admin	Manage doctors	High
Admin	Manage patients	High
Admin	Manage users	High
Admin	Manage appointment slots	High
Reports	Dashboard analytics	Medium
Reports	Generate reports	Medium
Security	Audit log	Medium
Security	Role-based access control	High
Future enhancement	Telemedicine consultation	Low